

360 Perspective

Private Infrastructure

April 2026

Important disclosures can be found in the Disclosures Appendix.



WS Global CIO Office

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What are Infrastructure Assets?

Infrastructure assets are the networks and systems that provide essential services, facilitate economic activity and enable the movement or storage of goods, water, energy, data and people



Traditional Infrastructure

Upgrades on roads, rails, transport network (e.g., ports, waterways and airports); modernizing public transit; build, preserve and retrofit civic and commercial real estate; upgrading aging water infrastructure including water treatment plants, pipes, sewers, water sanitation, improving access to clean water.



Renewable Infrastructure

Investments to accelerate the adoption of renewables in primary energy supply to meet Paris agreement commitments: Upgrading the electric grid, building electric vehicle charging stations, supporting sustainable farming and circular economy initiatives, decarbonizing the energy sector.



Digital Infrastructure

Investments in digitalization initiatives (cloud, 5G, data centers), expansion of high-speed broadband, funding of scientific research, granting or improving digital access to underserved segments of the population.

Source: Standard Chartered



Key characteristics of infrastructure assets

Key characteristics

Summary

'Real' Asset Class

- Infrastructure represents a tangible asset – i.e., a combination of land and structures that constitutes real property. Real property is expected to almost always retain a residual value, which is particularly attractive during periods of distress

Essential Services

- Given the essential nature of Infrastructure services, user demand tends to be relatively inelastic. Therefore, they tend to exhibit a lower correlation to the economic cycle compared with other sectors

High Barriers of Entry

- Infrastructure requires a high level of initial capital investment, which acts as a significant impediment to potential competitors entering the market. Assets can enjoy monopolistic or quasi-monopolistic market positioning

High Regulation

- Infrastructure assets are usually essential services providers which are important in the function of society and economy. Therefore, governments tend to exercise strict regulation over them. While regulations can vary, they usually make cash flows more stable

Diversified end-user base

- The counterparties to infrastructure assets are generally a widely diversified group of end users, which helps stabilize cash flows

Cash flow predictability

- Regulation provides long-term revenue visibility to infrastructure investments

Inflation hedge

- Depending on the type of infrastructure asset, price inflation can sometimes be passed on to the end consumer. Most regulatory frameworks allow regulated assets to use inflation-indexed user tariffs

Key
characteristics
of infrastructure
assets

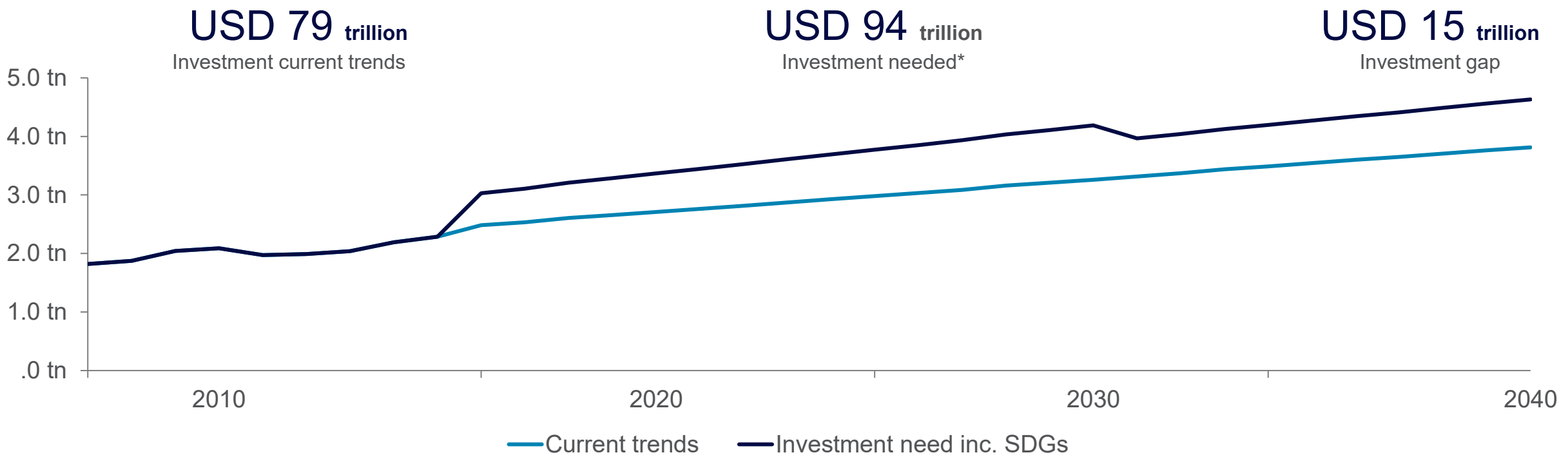
Source: Deutsche Asset Management, Standard Chartered



The Infrastructure “super-cycle”

Post-COVID, major governments are investing heavily in infrastructure assets to achieve supply chain resilience, digital independence and environmental targets

Global infrastructure investment spend has trailed behind actual needs



Source: Global Infrastructure Outlook, Standard Chartered
*Includes the additional investment needed for countries that have not yet met the SDGs
Note: values refer to estimated investment spending and needs until 2040
SDG refer to United Nations Sustainable Development Goals

The definition of infrastructure is evolving to meet future needs

From traditional, asset-heavy sectors...

Asset-heavy

Large; physical, high-capex-assets

Highly regulated

Government-owned and monopoly models

Linear and centralized

Centralized grids, fixed networks

Capex-intensive

Heavy investments upfront

Built on long-established technologies

Fossil fuels, resource-heavy

... to inclusion of services, technology, and distributed systems



Tech-enabled

Digital, software-driven infrastructure as a service



Market-driven

Private-sector competition and digital platforms



Decentralized and modular

Distributed, flexible, and demand driven



Opex and services

Pay-as-you-go, cloud-based subscription models

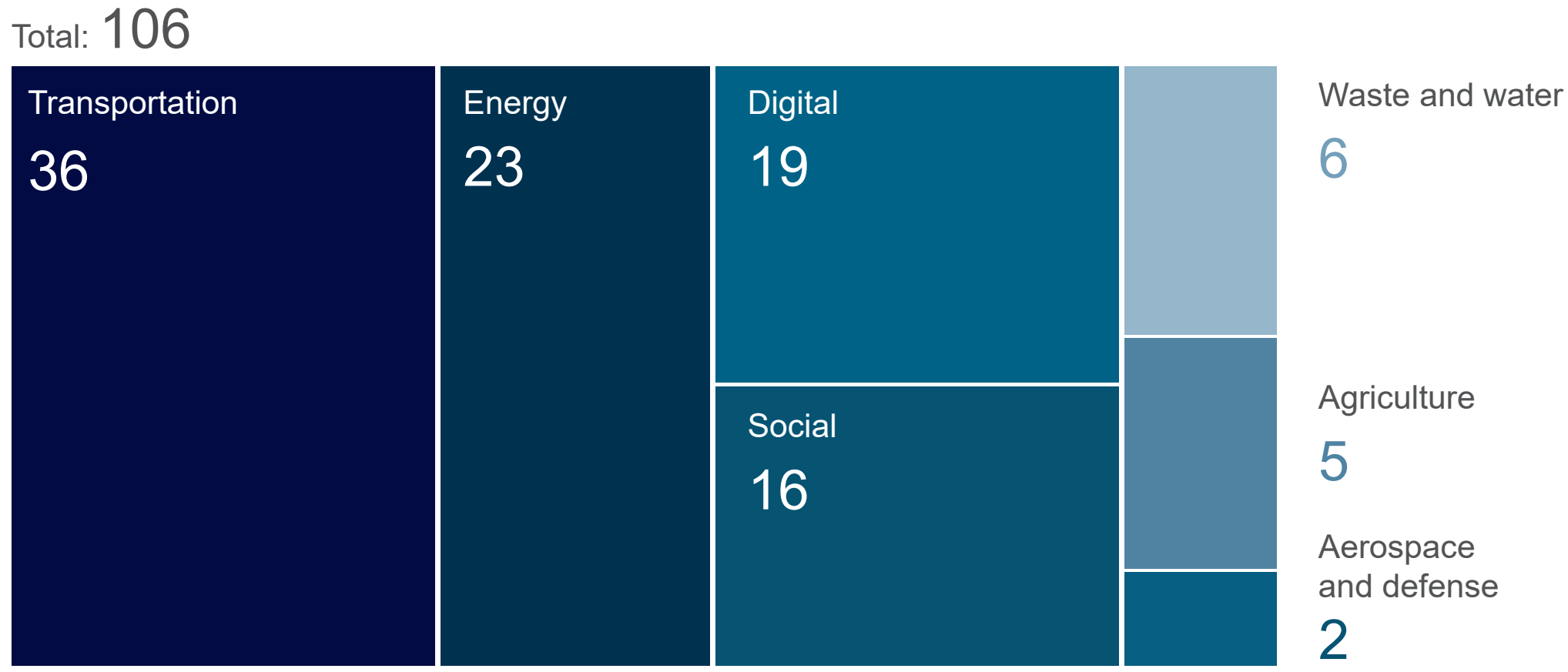


Built on scaling and disruptive technologies

Low-carbon, circular economy, sustainability driven

Cumulative infrastructure investment is expected to reach as high as \$106 trillion by 2040

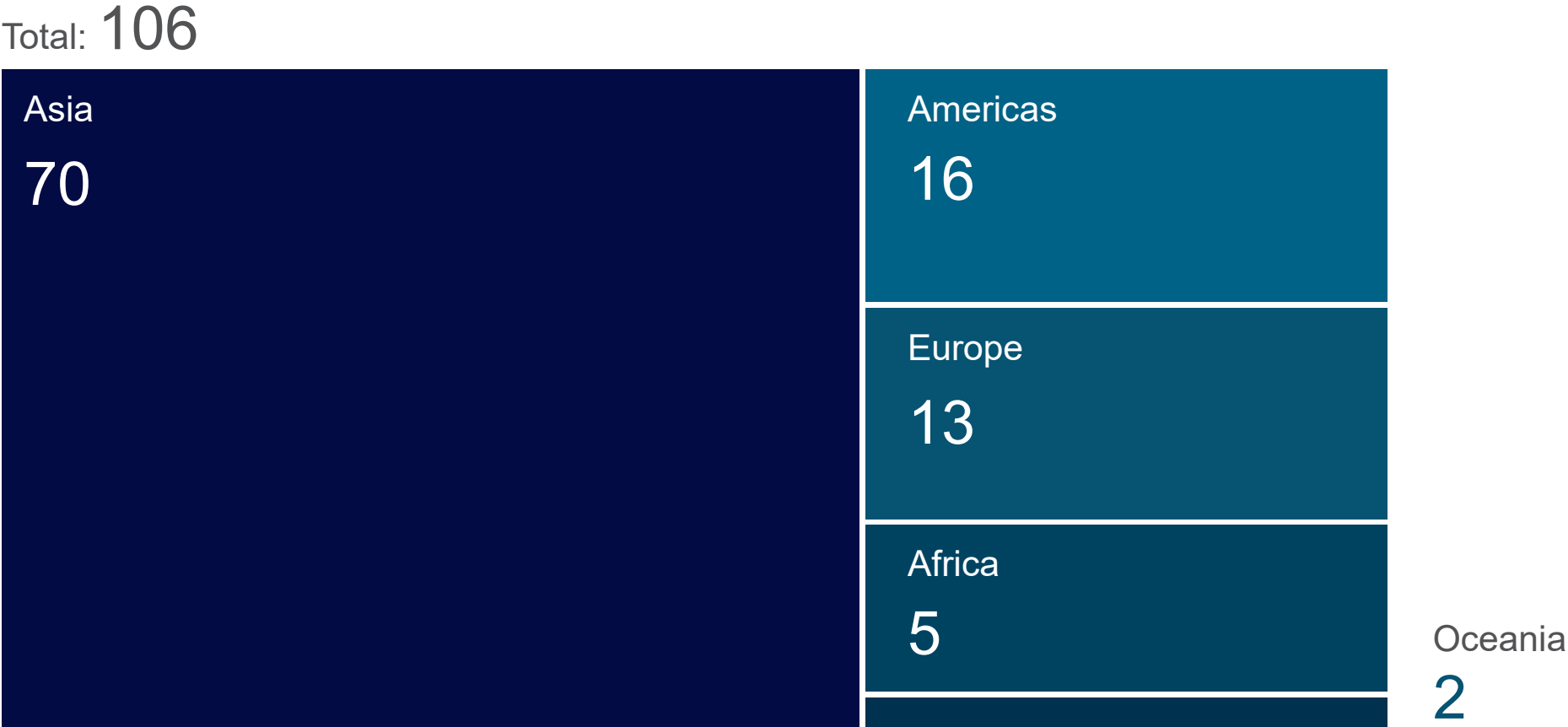
Total infrastructure investment projected through 2040, by sector, \$ trillion



Source: Food and Agriculture Organization; Global Infrastructure Hub; International Energy Agency; International Monetary Fund; Organisation for Economic Co-operation and Development; Preqin; United Nations; World Bank; World Economic Forum; McKinsey
Note: Figures do not sum, because of rounding

Asia could receive two-thirds of the total infrastructure investment through 2040

Total infrastructure investment projected through 2040, by region, \$ trillion



Source: Food and Agriculture Organization; Global Infrastructure Hub; International Energy Agency; International Monetary Fund; Organisation for Economic Co-operation and Development: Preqin; United Nations; World Bank; World Economic Forum; McKinsey

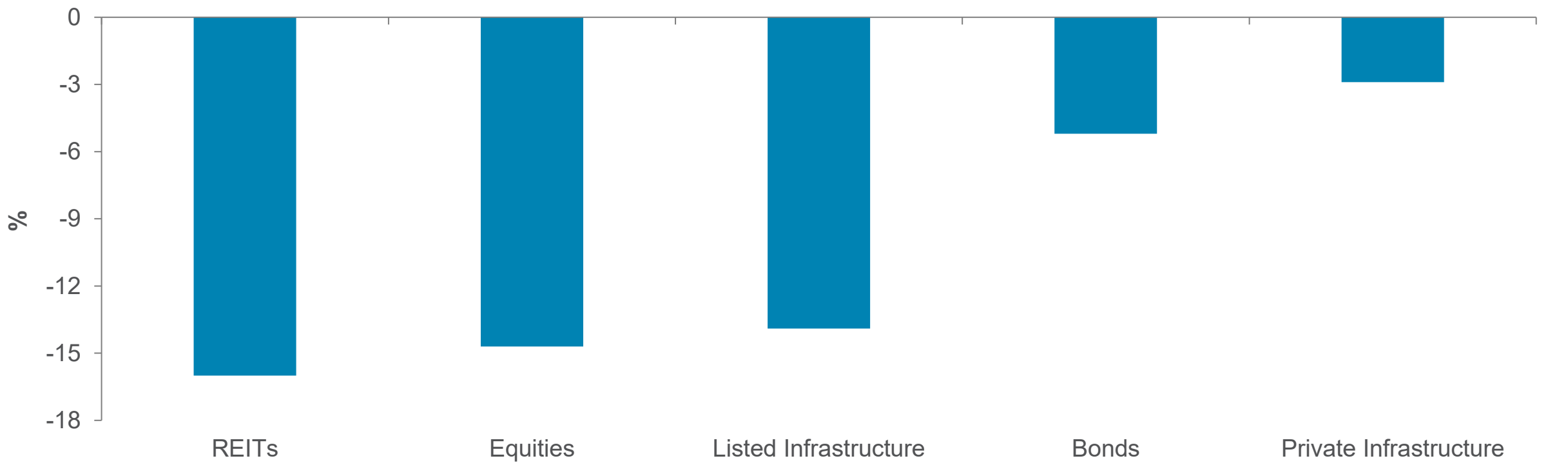


Portfolio Benefits of Private Infrastructure

Private infrastructure assets have demonstrated resilient returns during drawdowns

History suggests that infrastructure assets tend to perform relatively well compared to traditional assets during worst performing quarters for equity markets

Average return by asset class during 10 worst quarters for equity markets



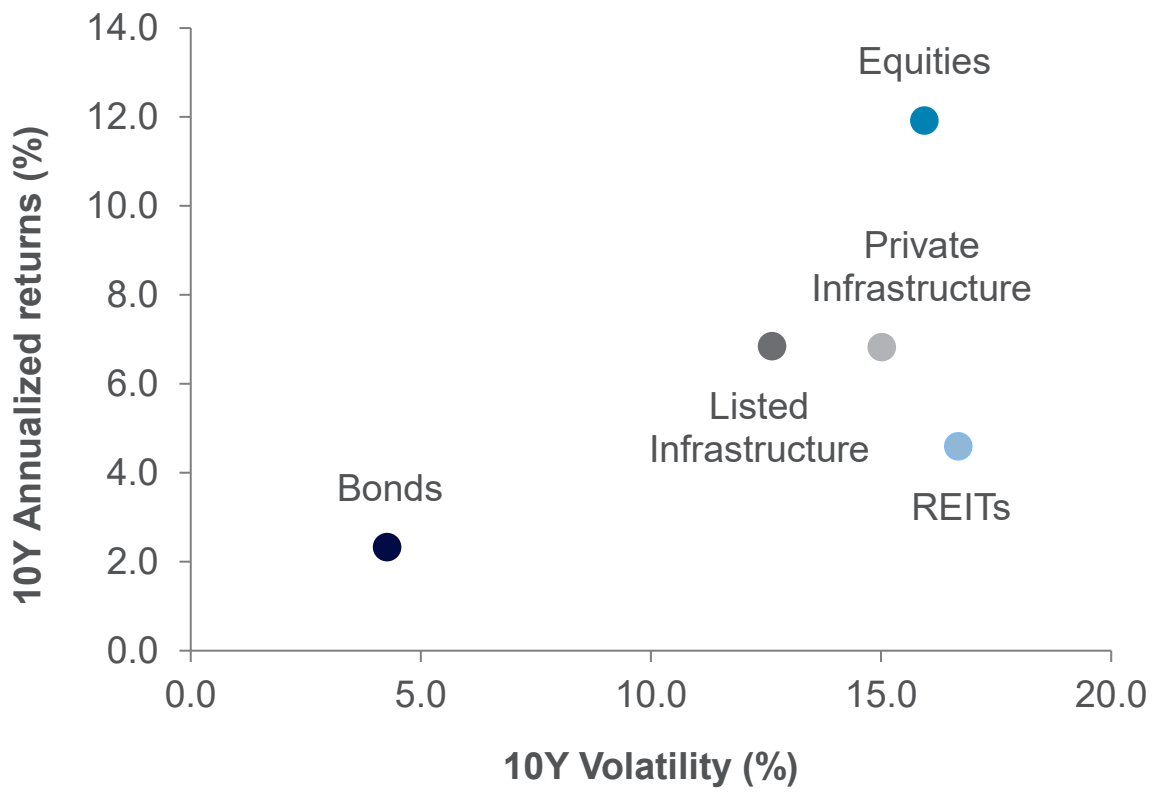
Source: Allianz Research, Standard Chartered



Private infrastructure is often less correlated to traditional asset classes

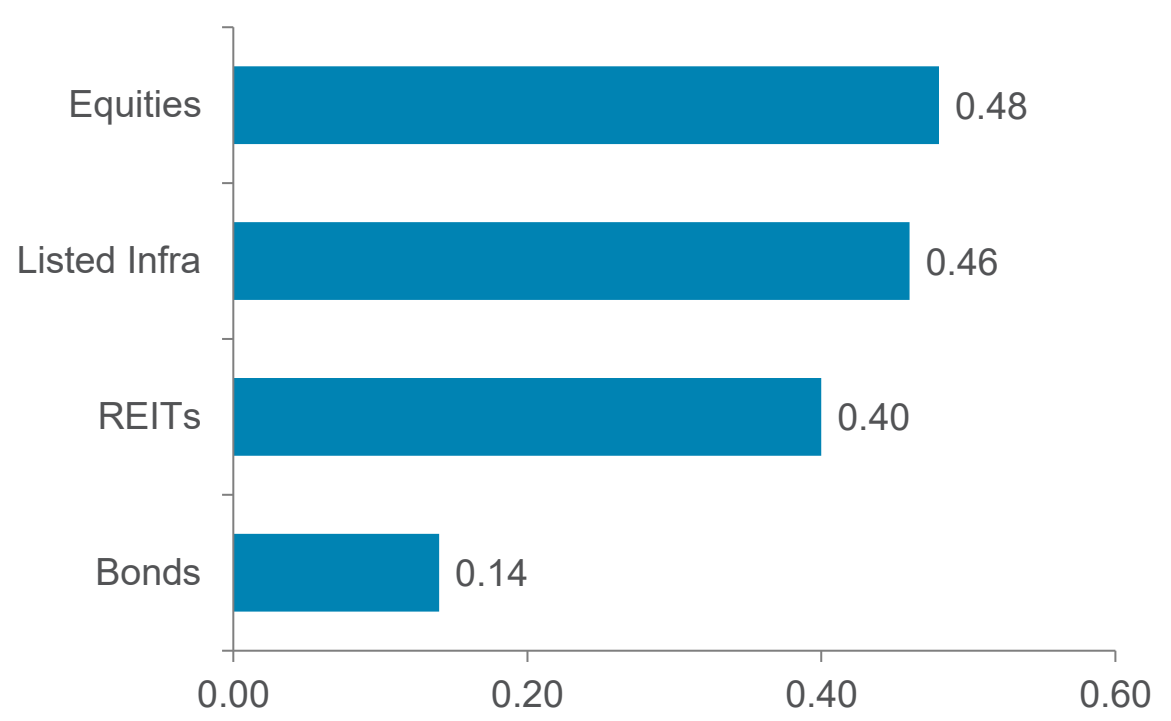
It also has relatively lower volatility compared to equities and REITs, given its regular cash flow which can mitigate downside price volatility

10Y Risk/Return matrix across Asset Class



Source: Bloomberg, The Scientific Infrastructure & Private Assets Research Institute (EIPA), Standard Chartered Bank
Data as of September 2025

10Y Correlation with Private Infrastructure



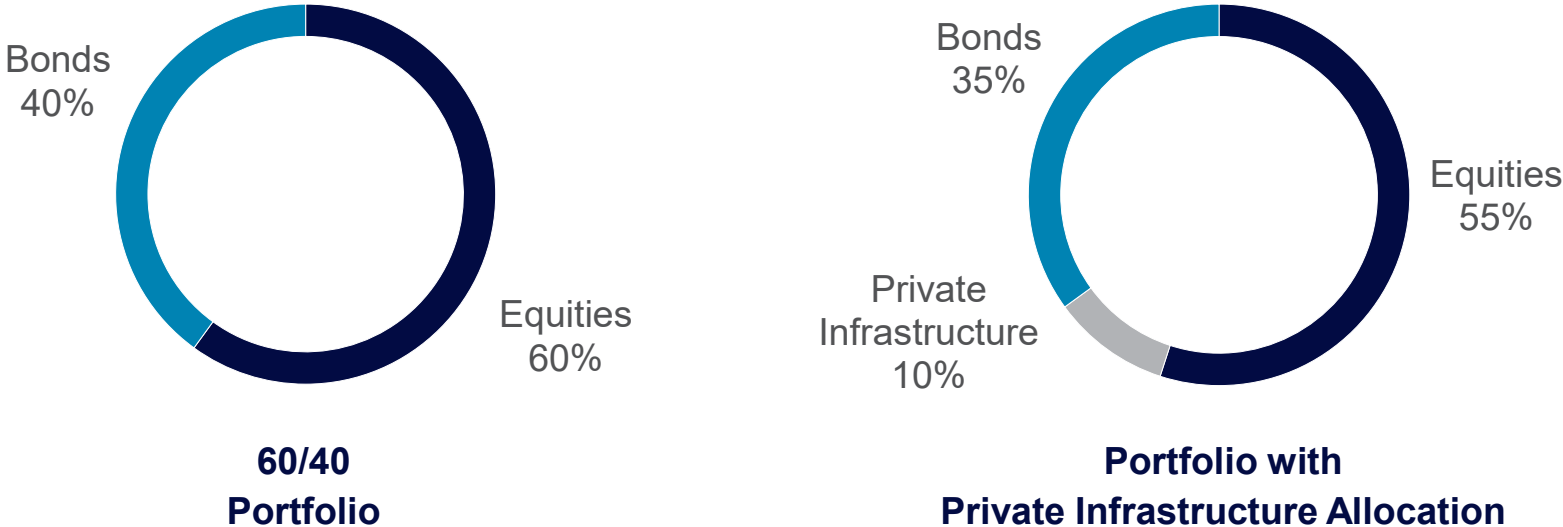
Source: Allianz Research, Standard Chartered Bank
Data as of 2024



Adding Private infrastructure could improve risk-adjusted returns in a portfolio

An allocation towards Private Infrastructure, which is a less volatile asset, can improve a portfolio's risk-adjusted returns

Performance of a portfolio with a private infrastructure allocation vs a Traditional 60/40 Portfolio



	60/40 Portfolio	Portfolio with Private Infrastructure Allocation
Historical 10Y Ann. Return	8.7%	9.0%
Historical Volatility	11.1%	10.3%
Sharpe Ratio	0.47	0.54

Source: StashAway, Standard Chartered
Data as of September 2024



Infrastructure assets are generally resilient through an economic cycle

Private Infrastructure assets can help hedge against inflation as cash flows are typically inflation-linked, which is a valuable attribute during periods of high inflation



Income

Given their essential nature, infrastructure investments can generate relatively stable and resilient cash-flows in a range of market environments



Downside Protection

The essential nature of the business allows it to remain resilient through an economic cycle, usually limiting drawdowns at least relative to public market assets



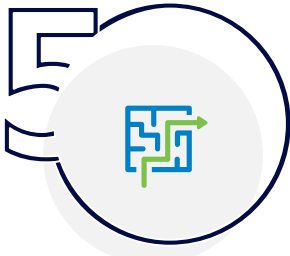
Stability of Cashflow

Infrastructure's contractual business nature helps provide investors with long-term visibility into cash flows with usually less exposure to price volatility and volume risk



Inflation Protection

Private Infrastructure cash flow are usually indexed to inflation. There are also often pass-through clauses for higher costs given the essential nature of their underlying business



Diversification

Private Infrastructure has a relatively low volatility relative to, and correlation with, traditional asset classes



Sustainability

The global shift towards net-zero suggests that investment into supporting infrastructure will increase as countries push to overhaul their existing carbon-based infrastructure

Source: Standard Chartered

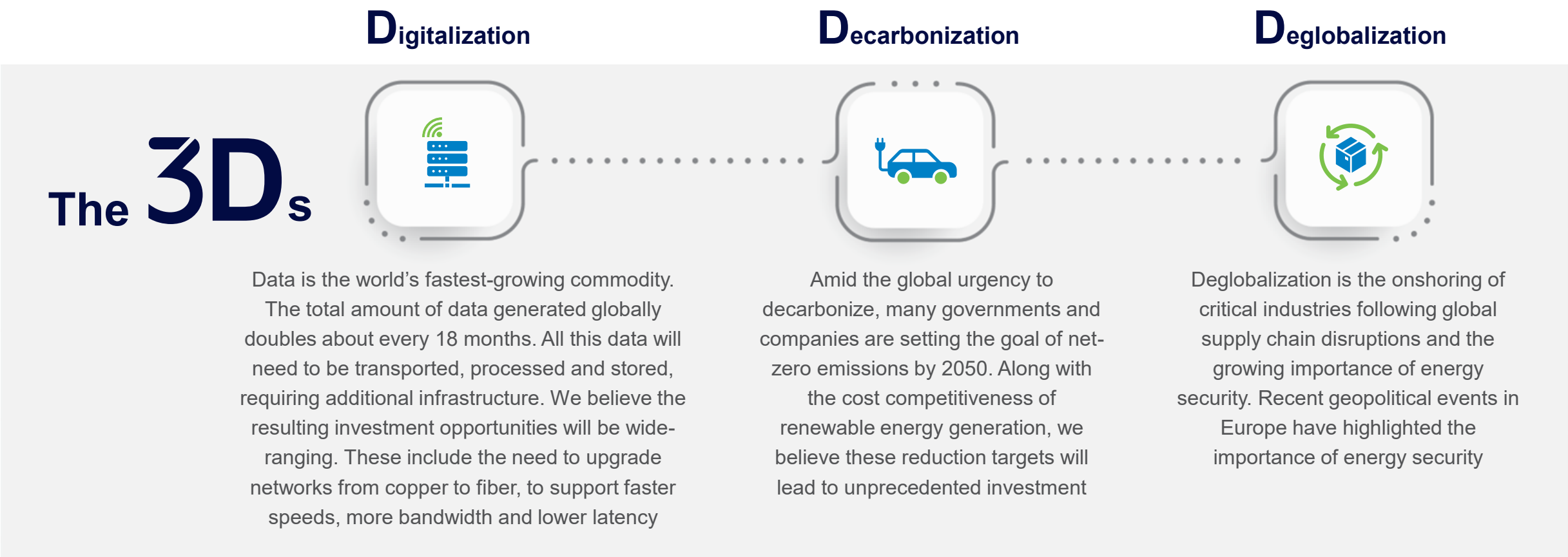




Key Drivers of Infrastructure spending

The 3Ds of Infrastructure Opportunities

The infrastructure needs to enable long-term trends, including the investment to meet the staggering growth in data consumption, the shift away from traditional fossil fuels and the acceleration toward e-commerce, require a tremendous amount of capital

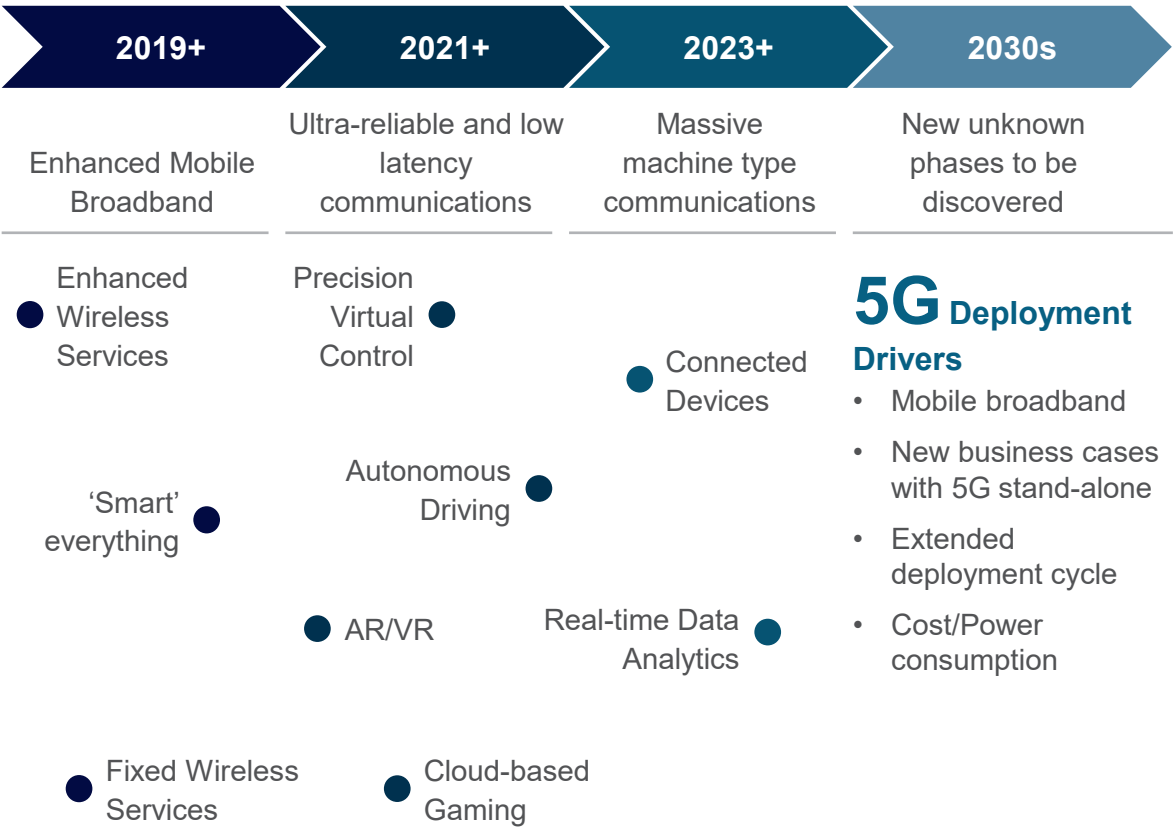


Source: Brookfield, Standard Chartered

Greater 5G adoption to drive spending on digital infrastructure

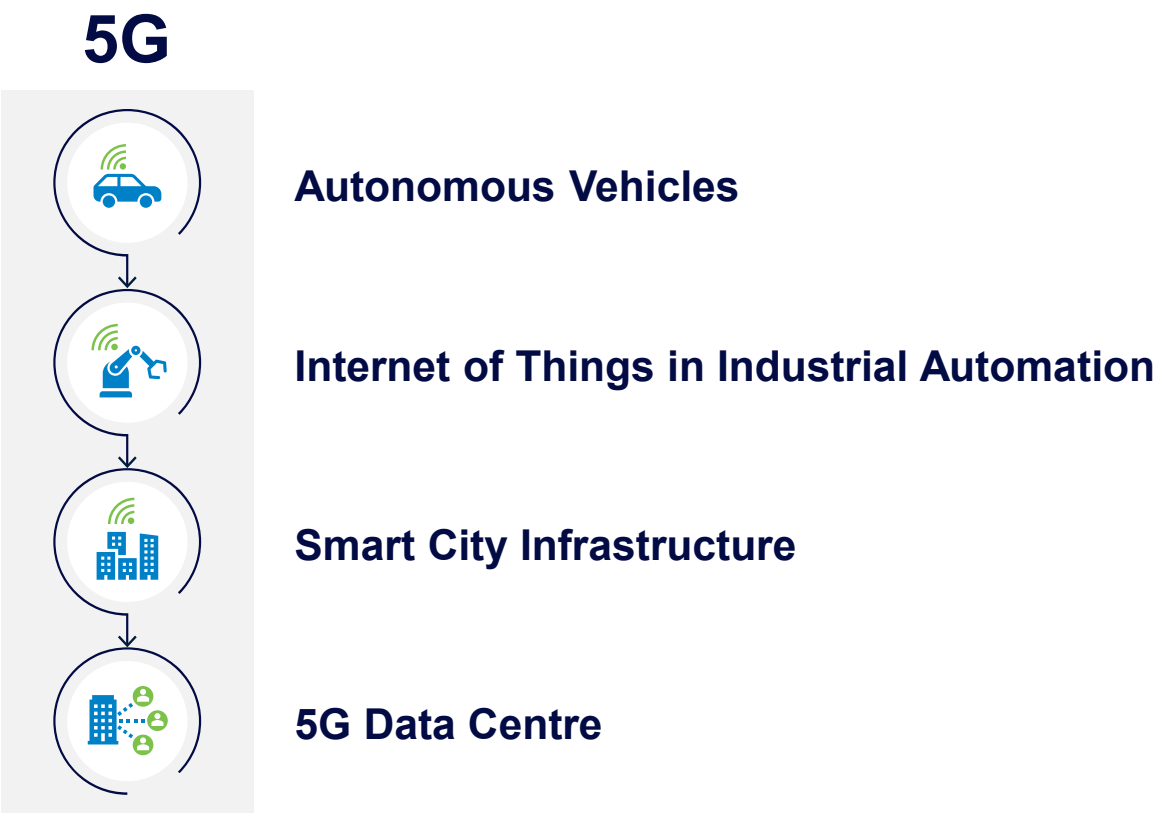
The construction of higher speed, lower latency 5G networks is expected to open a host of applications that previously were not viable. These applications are likely to drive Digital Infrastructure spending

5G Development Stages



Source: NXP Semiconductor, Standard Chartered

Infrastructure needs to support the Application of 5G

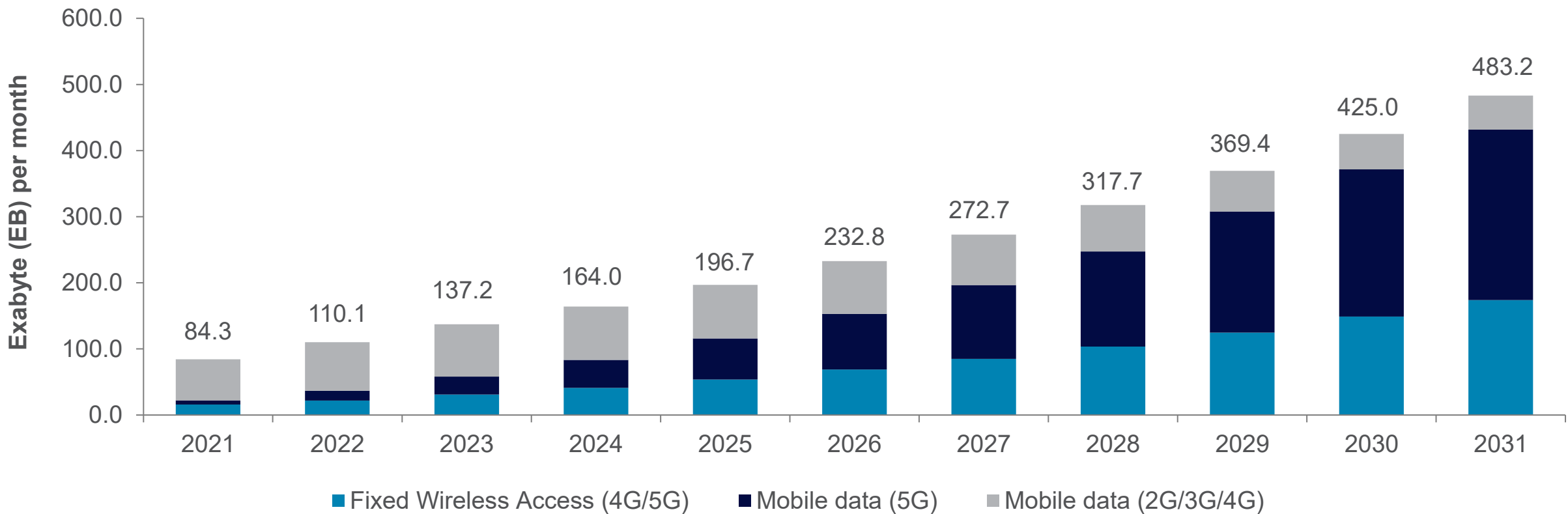


Source: Digi, Standard Chartered

The future of Data Consumption

Mobile data network consumption is expected to grow steadily and the need for more data centers and faster network infrastructure will drive investment needs

Global mobile network data traffic and forecasts



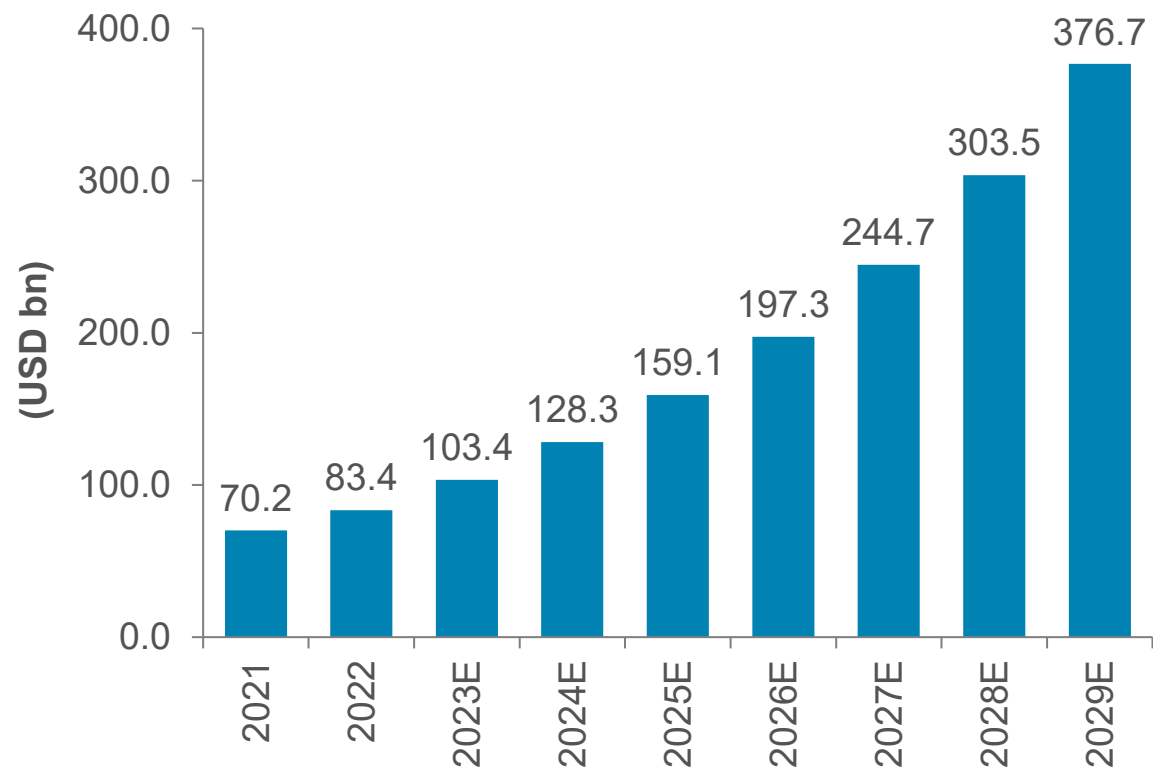
Source: Ericsson Mobility Report, Standard Chartered



The future of Digitalization

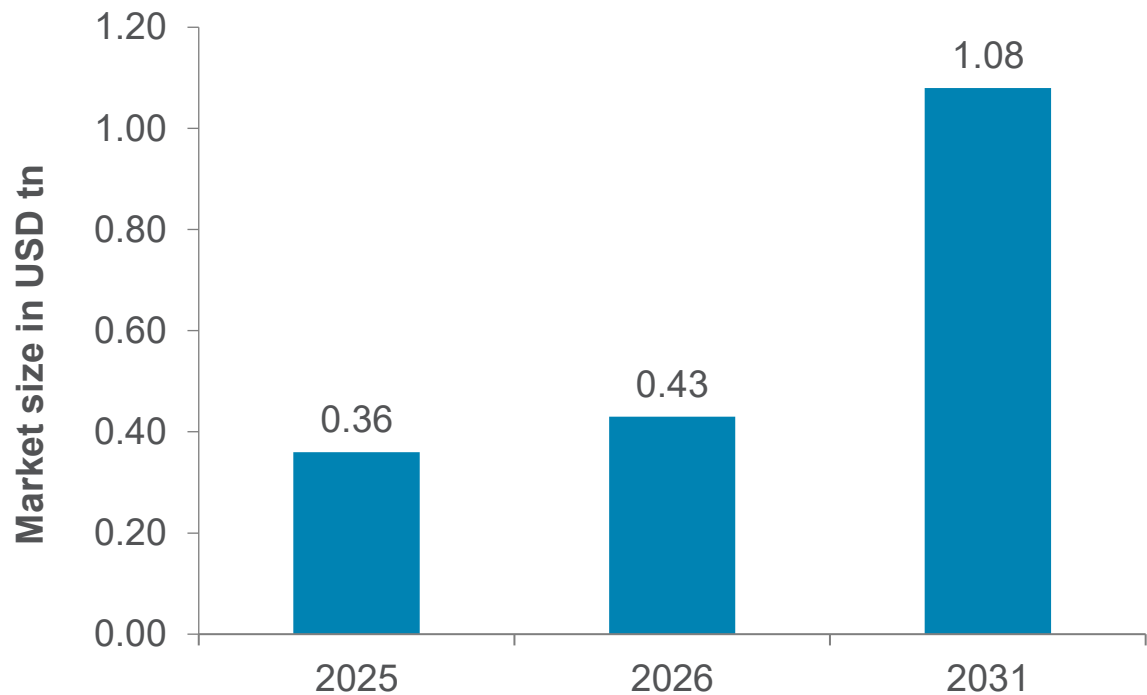
As the world marches towards digitalization, the growth in Digital Infrastructure spending is set to persist

Growth of Cloud Storage Market size



Source: Upwork, Synergy Research Group, Global X, Statista, Standard Chartered

Digital infrastructure markets are expected to grow at a CAGR of 20% from 2026 to 2031



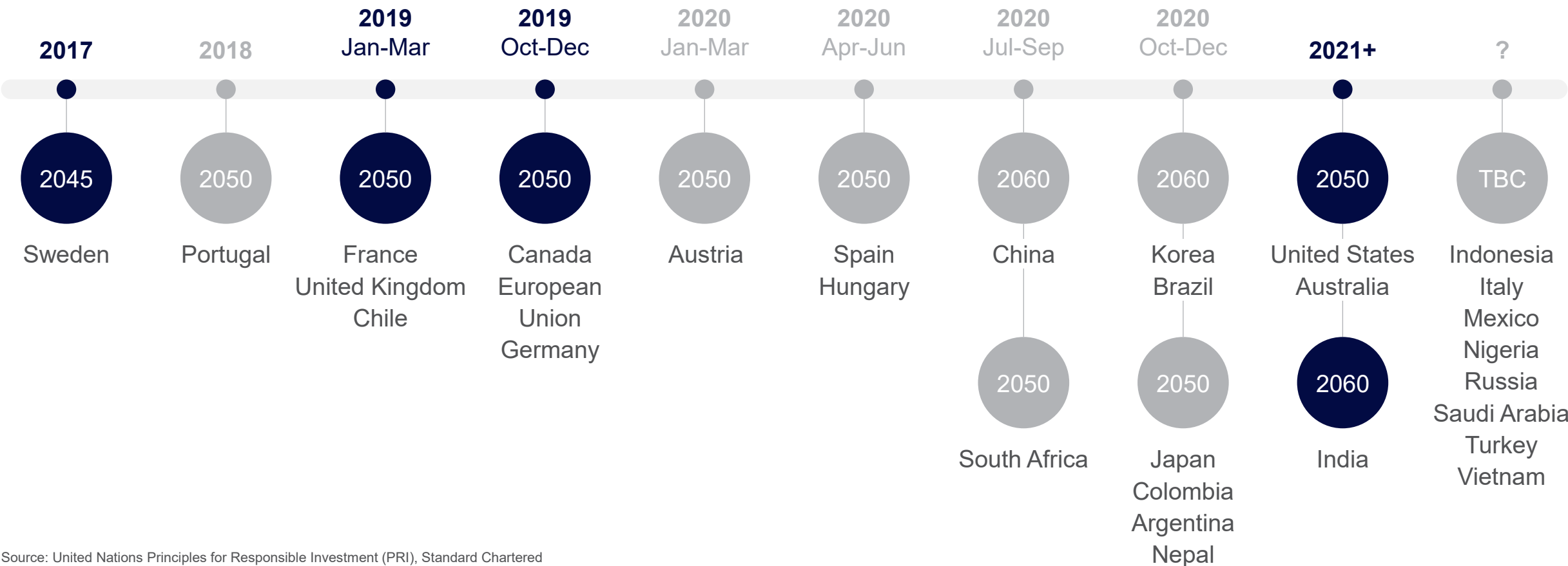
Source: Mordor Intelligence, Standard Chartered



Government pledges toward net zero carbon emissions

As the world look towards decarbonizing in the decades ahead, policies and investments are needed to overhaul existing infrastructure to achieve net-zero targets

Announcement timeline for net zero emissions targets



Source: United Nations Principles for Responsible Investment (PRI), Standard Chartered



Unprecedented investment in the build-out of renewable infrastructure

Many reduction targets will require significant capital investment to overhaul the power generation sector, the world's largest source of carbon emissions

Expected investment needed to transition energy systems by 2050

USD 150 trillion

Total investment to transition energy system by 2050



Build out renewable
power generation



Global
Electrification



Enhance electrical
infrastructure



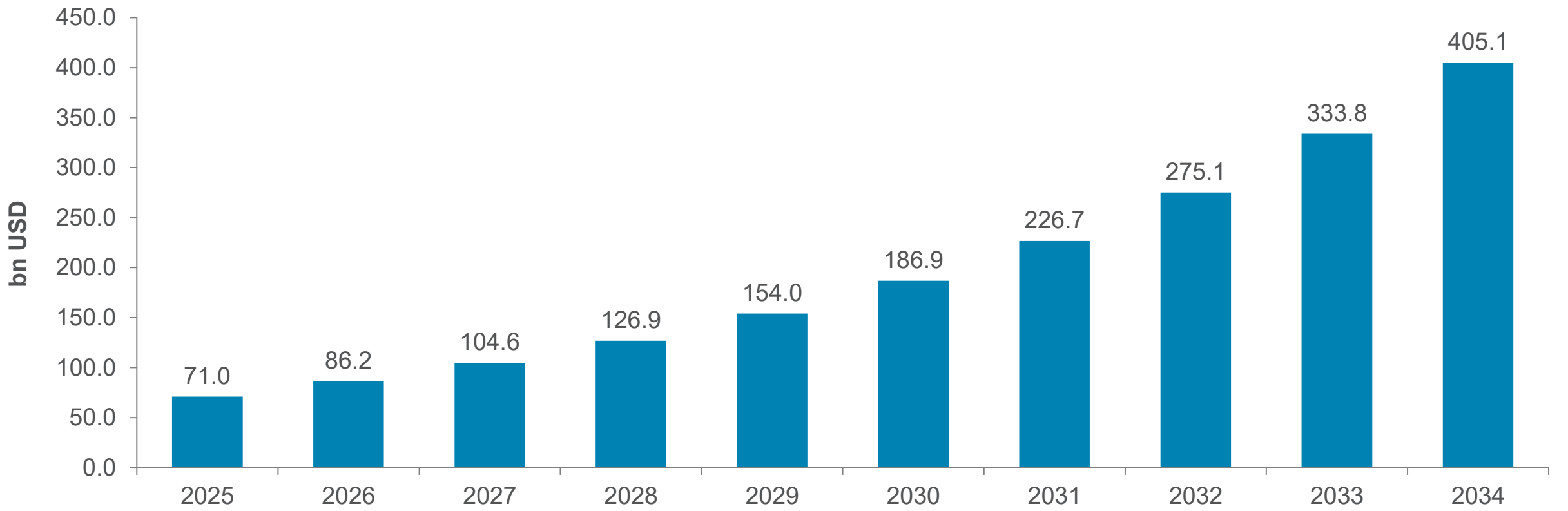
Repurpose existing
infrastructure assets

Source: Brookfield, Standard Chartered

The renewable infrastructure rush

Rising environmental awareness and government pressure for carbon-neutral construction methods are driving market growth for sustainable infrastructure

Sustainable Infrastructure Market Size



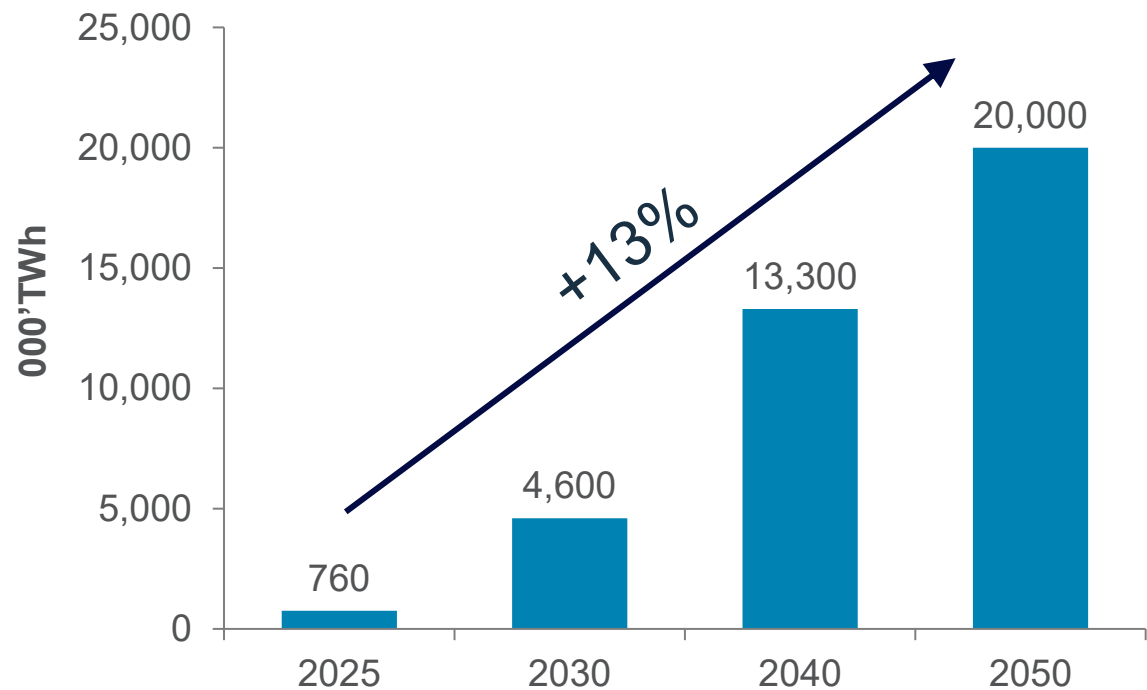
Source: Precedence Research, Standard Chartered



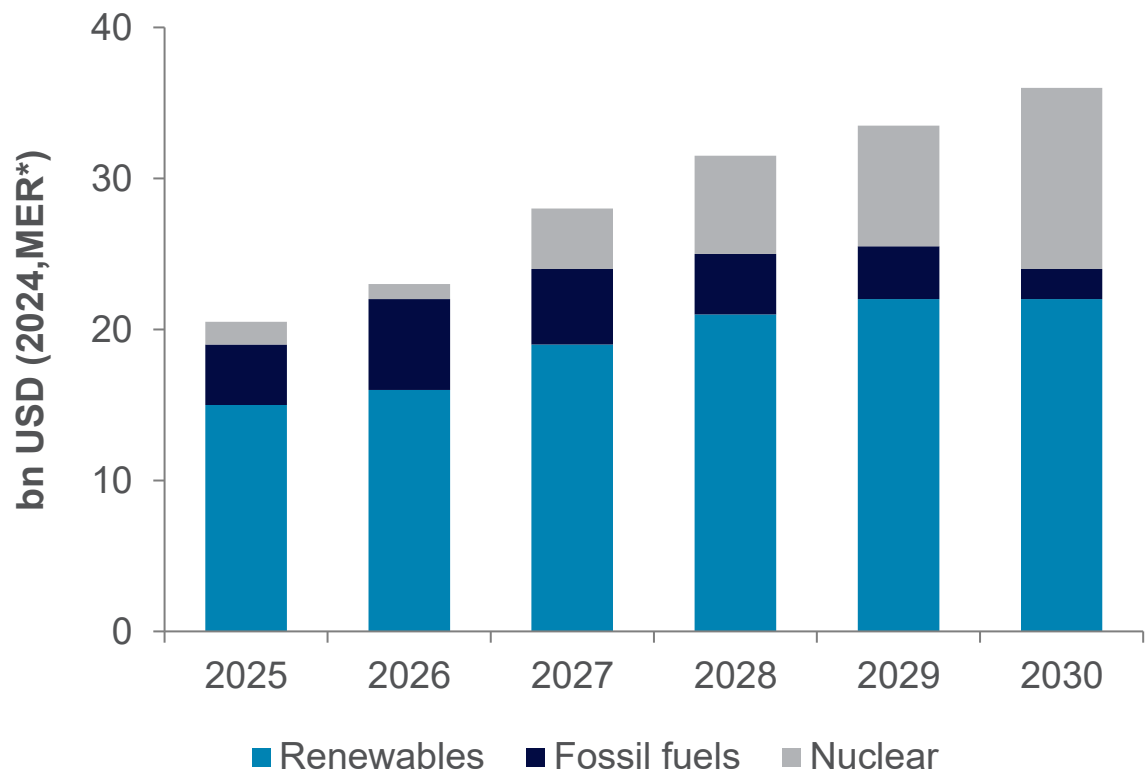
Acceleration of demand for global electricity

Data centres are driving demand for energy, and renewable energy infrastructure is meeting it

Global Electricity Demand Growth, with data centers projected 17% annual growth between 2025 to 2050



New power generation investment for data centers



Source: Brookfield, Standard Chartered

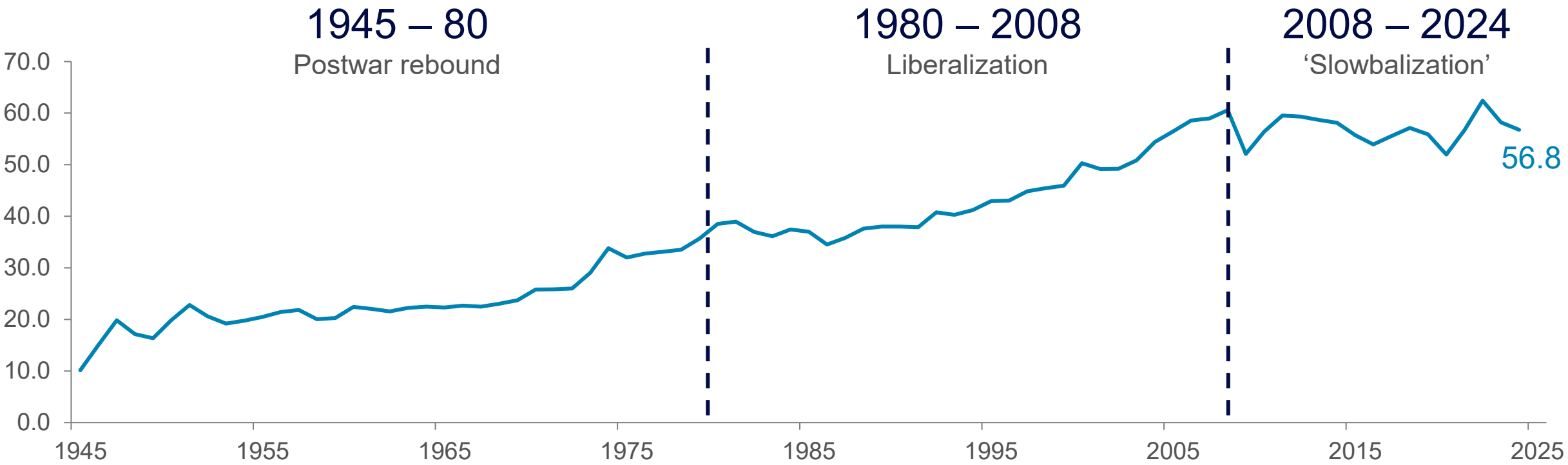
Source: International Energy Agency (IEA) World Investment Energy 2025 Report, Standard Chartered
* Market Exchange Rate



Globalization is in retreat for the first time since World War II

The pandemic has exposed the need for countries to onshore critical industries like energy security, supply chains and technology infrastructure

Global economy trade openness index

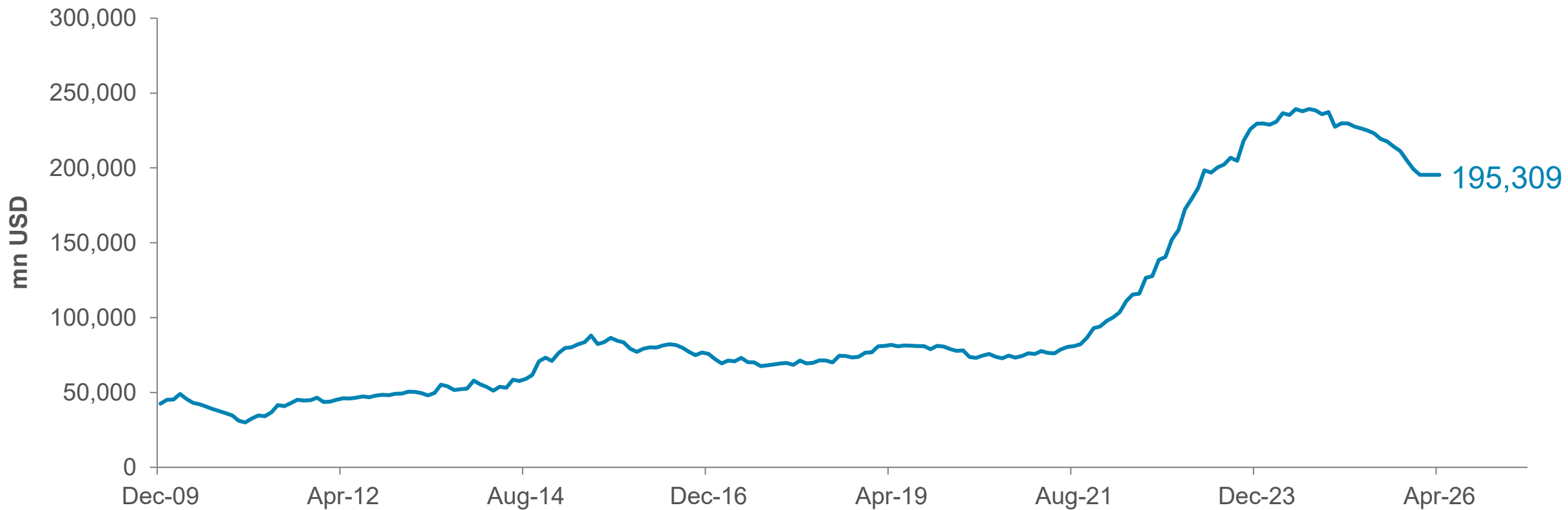


Source: Our World in Data, Peterson Institute for International Economics (PIIE), Standard Chartered
Note: National statistical organizations and central banks, OECD national accounts, and World Bank staff estimates (2026)Feenstra et al. - Penn World Table (2025)Klasing, Mariko J. and Milionis, P. (2014). "Quantifying the Evolution of World Trade, 1870-1949". Journal of International Economics 92(1), pp. 185-197Estevadeordal, A., Frantz, B., & Taylor, A. (2003). The Rise and Fall of World Trade, 1870-1939. The Quarterly Journal of Economics, 118(2), 359-407. Retrieved from <http://www.jstor.org/stable/25053910> – with major processing by Our World in Data

Rising geopolitical tension a threat to globalization

Geopolitical tensions is one of the drivers expected to drive reshoring efforts

Total US Private Construction Spending on Manufacturing



Source: Bloomberg, Standard Chartered



Takeaways: Decarbonization, Deglobalization and Digitalization to support spending on infrastructure assets

Key Drivers



Digitalization

Governments have earmarked digital infrastructure as a key focus area in their stimulus plans. Digital infrastructure is the foundation on which digital economic activities, technological applications and significant productivity gains can be built on.

Decarbonization

A convergence of interest in climate change from governments, corporates and individuals is accelerating the transition to a 'net zero' economy, of which the power and transport sector have a key role to play.

Deglobalization

Recent geopolitical events have highlighted the importance of onshoring critical industries. The pandemic has also revealed the inefficiencies and un-reliabilities in the existing supply chain infrastructure.



Key Risks

Key Risks

Risks	Summary
01 Political and Regulatory Risk	• Infrastructure assets are essential services that ensures the functioning of society and the modern economy; hence governments tend to maintain certain control or regulation over them • The level of exposure to political developments as well as the tightness of regulation differs from one country or sector to another and can have a significant impact on investment returns
02 Construction Risk	• In greenfield projects, construction risk are involved. Such risk includes project overruns and delays. These risks can be common for complex projects such as bridges or energy plants. Any project overruns or delays will usually cause delayed revenues and reduced returns
03 Operational Risk	• When an infrastructure asset is operating, operational risks will be involved, including performance or maintenance risks. Performance risk can include falling short of volume or price objectives or increased costs of doing business. Maintenance risk includes unplanned maintenance costs that can reduce operating cash flow levels. Exposure to operational risk varies by asset type • Regulated assets like electricity grids are typically more protected against certain risks through regulatory formulas or cost pass through to the customer, while unregulated assets are usually more exposed
04 Financial Risk	• Depending on the level of leverage and the debt structure, infrastructure assets can be exposed to interest rate volatility or refinancing risk. If not hedged, interest costs may increase, reducing operating cash flow levels

Source: Deutsche Asset Management, Standard Chartered



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